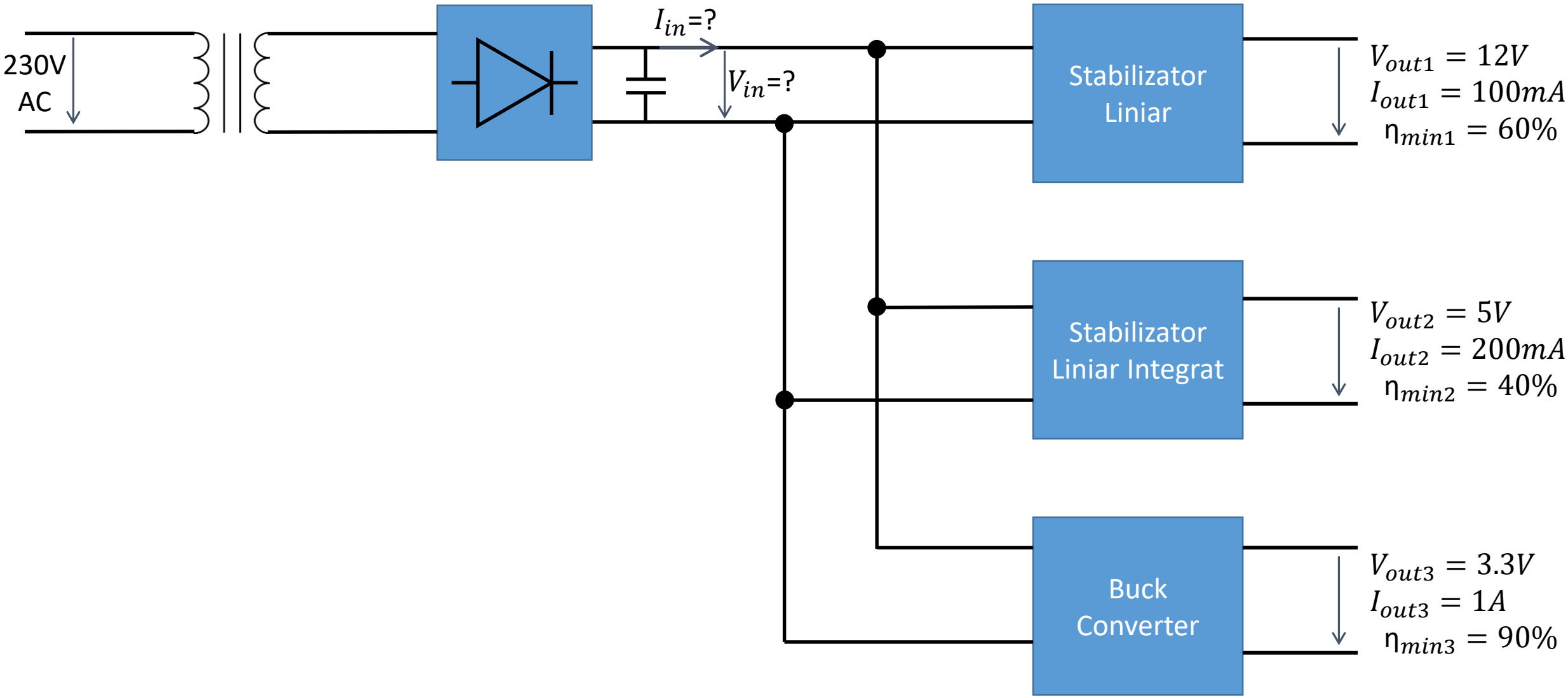
Analiza și Dimensionarea unui Etaj de Alimentare

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An IV, gr. 2142


230V
AC
I_{in}=?
V_{in}=?
Stabilizator
Liniar
$V_{out1} = 12V$
$I_{out1} = 100mA$
$\eta_{min1} = 60\%$
Stabilizator
Liniar Integrat
$V_{out2} = 5V$
$I_{out2} = 200mA$
$\eta_{min2} = 40\%$
Buck
Converter
$V_{out3} = 3.3V$
$I_{out3} = 1A$
$\eta_{min3} = 90\%$

$$P_{out1} = V_{out1} * I_{out1}$$
$$= 12V * 100mA$$
$$= 1.2W$$
$$\eta_{min1} = \frac{P_{out1}}{P_{in1}} = 60\%$$
$$\Rightarrow P_{in1_max} = \frac{P_{out1}}{0.6} = 2W$$
$$P_{in1} \in (1.2 ; 2]\ W$$

$$P_{out2} = V_{out2} * I_{out2}$$
$$= 5V * 200mA$$
$$= 1W$$
$$\eta_{min2} = \frac{P_{out2}}{P_{in2}} = 40\%$$
$$\Rightarrow P_{in2_max} = \frac{P_{out2}}{0.4} = 2.5W$$
$$P_{in2} \in (1 ; 2.5]\ W$$

$$P_{out3} = V_{out3} * I_{out3}$$
$$= 3.3V * 1A$$
$$= 3.3W$$
$$\eta_{min3} = \frac{P_{out3}}{P_{in3}} = 90\%$$
$$\Rightarrow P_{in3_max} = \frac{P_{out3}}{0.9} = 3.66W$$
$$P_{in3} \in (3.3 ; 3.66]\ W$$

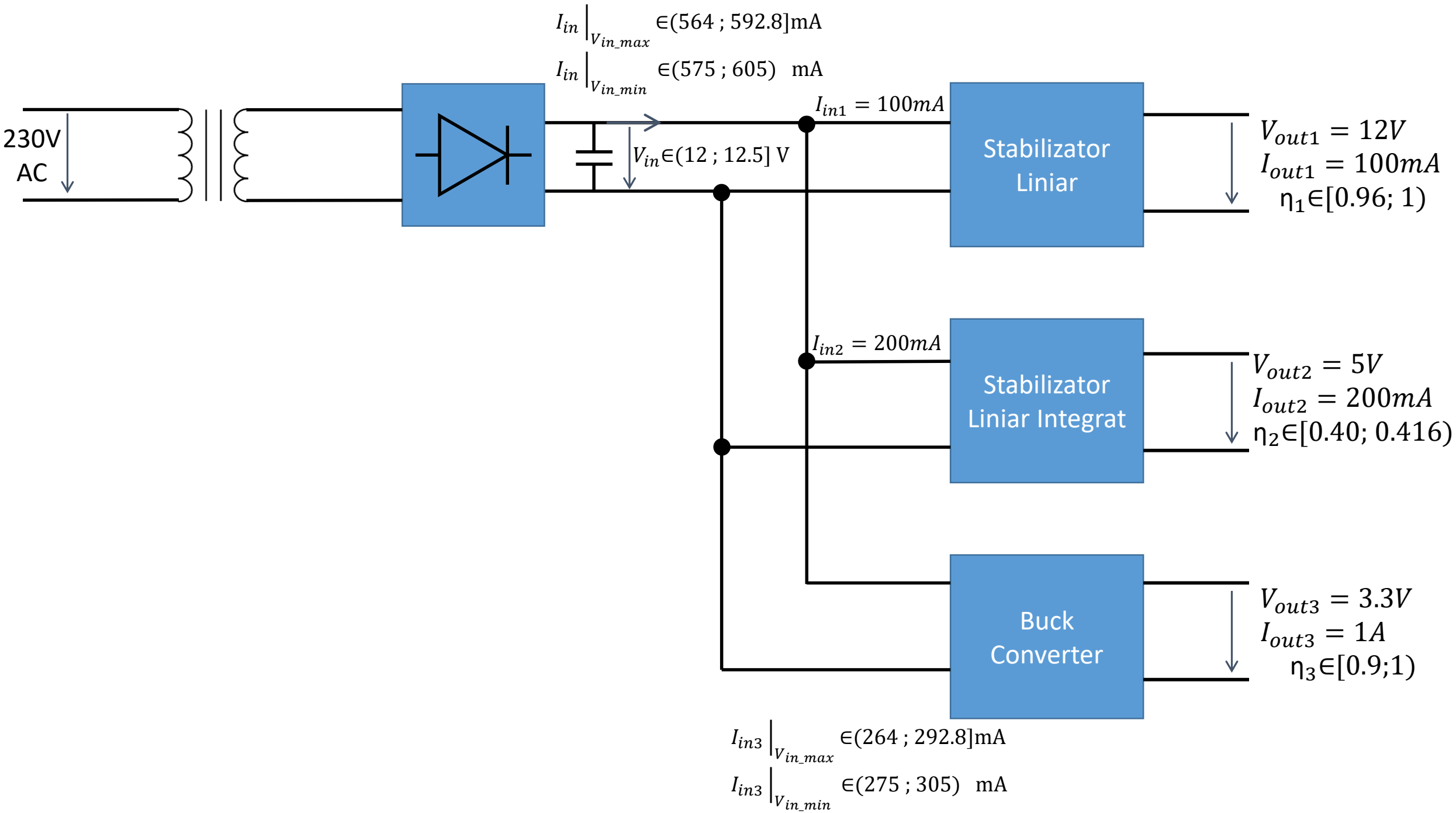
$$I_{in_stabilizator_liniar} = I_{out_stabilizator_liniar}$$

$$I_{in1} = I_{out1} = 100mA$$
$$P_{in1} = V_{in} * I_{in1}$$
$$P_{in1} \in (1.2 ; 2]\ W$$
$$\Rightarrow V_{in} \in (12 ; 20]\ V$$

$$I_{in2} = I_{out2} = 200mA$$
$$P_{in2} = V_{in} * I_{in2}$$
$$P_{in2} \in (1 ; 2.5]\ W$$
$$\Rightarrow V_{in} \in (5 ; 12.5]\ V$$

$$\Rightarrow V_{in} \in (12 ; 12.5]\ V$$
$$P_{in3} \in (3.3 ; 3.66]\ W$$
$$I_{in3} = \frac{P_{in3}}{V_{in}}$$
$$I_{in3}\Big|_{V_{in_max}} \in (264 ; 292.8]mA$$
$$I_{in3}\Big|_{V_{in_min}} \in (275 ; 305)\ mA$$

$$I_{in} = I_{in1} + I_{in2} + I_{in3}$$
$$I_{in} = 300mA + I_{in3}$$
$$I_{in}\Big|_{V_{in_max}} \in (564 ; 592.8]mA$$
$$I_{in}\Big|_{V_{in_min}} \in (575 ; 605)\ mA$$


$I_{in}\big|_{V_{in_max}}$ ∈(564 ; 592.8]mA
$I_{in}\big|_{V_{in_min}}$ ∈(575 ; 605) mA
230V
AC
V_{in}∈(12 ; 12.5] V
$I_{in1} = 100mA$
Stabilizator
Liniar
$V_{out1} = 12V$
$I_{out1} = 100mA$
η_1∈[0.96; 1)
$I_{in2} = 200mA$
Stabilizator
Liniar Integrat
$V_{out2} = 5V$
$I_{out2} = 200mA$
η_2∈[0.40; 0.416)
Buck
Converter
$V_{out3} = 3.3V$
$I_{out3} = 1A$
η_3∈[0.9;1)
$I_{in3}\big|_{V_{in_max}}$ ∈(264 ; 292.8]mA
$I_{in3}\big|_{V_{in_min}}$ ∈(275 ; 305) mA

Vă mulțumesc pentru atenție!